Physics IGCSE	Summary formula	Physics grade 8
Content	**Symbols**	**System international unit (SI unit)**
1. Work done in straight line **2.** Work done by lift any objects over pully **3.** Work done with any angle respect with x-axis **4.** Electric power **5.** Instantaneous power **6.** Energy transfer **7.** Kinetic energy **8.** Potential energy **9.** Conservation of energy **10.** Efficiency **11.** Stopping distance	1. $W = Fd$ 2. $W = mgh$ 3. $W = Fd\cos\theta$ 4. $P = \frac{W}{t} = \frac{E}{t}$ 5. $P = Fv$ 6. $E = QV, E = Pt$ 7. $E_k = \frac{1}{2}mv^2$ 8. $E_p = mgh$ 9. $\frac{1}{2}mv^2 = mgh \Rightarrow v = \sqrt{2hg}$ 10. $eff = \frac{output}{input} \times 100\%$ 11. $d = d_r + d_b, d_r = v_r t$	Where, W is the work done measured in (J). d is the distance measured in (m). F is the force measured in (N). m is the mass of an object measured in (kg). θ is the angle measured in $(^\circ)$. P is the electric power measured in (W). t is the time taken measured in (s). v is the velocity measured in (m/s). E is the energy transfer measured in (J). Q is the electric charge measured in (C). **Noted:** $1kW = 1000W, 1hp = 1cp = 750W$ $1MW = 1000000W = 1\times10^6 W$ E_k, E_P are the kinetic and potential energy measured in (J). h is the height measured in (m). d_r is the reaction distance measured (m).
12. Density of substance **13.** Volume of rectangular block and area of circle **14.** Pressure **15.** Hydraulic machine **16.** Pressure depend on Depth **17.** Pressure in U-tube manometer. **18.** Upthrust force	12. $\rho = \frac{m}{V}, m = \rho V$ 13. $V = L\times W\times h, A = \pi r^2, A = \frac{\pi d^2}{4}$ 14. $P = \frac{F}{A}$ 15. $F = \frac{f}{a}\times A = f\times\frac{A}{a}$ 16. $P = \rho gh$ 17. $P = P_{atm} + \rho gh, P_{atm} = 1.03\times10^5 Pa$ 18. *Upthurst* $force = \rho g(h_2 - h_1)A$	ρ is the density measured in (kg/m^3). V is the volume measured in (m^3). P is the pressure measured in (Pa). A is the cross-sectional area measured in (m^2). f is the force act on small piston in (N). F is the force act on large piston in (N). a is the area of small piston measured in (m^2). A is the area of large piston measured in (m^2). g is the gravitational field measured (m/s^2). P_{atm} is atmosphere pressure measured (Pa).
19. Electric power **20.** Energy transfer **21.** Cost of using electricity	19. $P = \frac{W}{t} = \frac{E}{t}, P = VI = \frac{V^2}{R} = RI^2$ 20. $W = E = VIt = \frac{V^2}{R}t = RI^2t$ 21. $\cos t = E(kWh)\times price/kWh$	P is the power electric measured in (W). R is the resistance measured in (Ω). I is the electric current measured in (A). $E(kWh)$ is the energy measured in (kWh). T is the period measured in (s).
22. Speed of sound **23.** Speed of echo sound **24.** Period and frequency relation	22. $v = \frac{d}{t}, v = \lambda f, \lambda = \frac{v}{f}$ 23. $v = \frac{2d}{t}$, 24. $T = \frac{1}{f}, f = \frac{1}{T}$	v is the speed of sound measured in (m/s). d is the distance measured in (m). λ is the wavelength measured in (m). f is the frequency measured in (Hz).

REVIEW FOR FINAL EXAM GRADE 8

A. Multiple choice questions

1. Which of the following is the formula of electric power?
a. $P = E \times t$ **b.** $P = V \times R$
c. $P = VI$ **d.** $P = \frac{V}{R}$

2. Which of the following is the energy transfer?
a. $E = P \times t$ **b.** $E = P + t$
c. $E = \frac{P}{t}$ **d.** $E = VI^2t$

3. Which of the following is the force per unit area?
a. Work done **b.** Pressure
c. Power **d.** Density

4. Which of the following is the formula of pressure?
a. $P = F \times A$ **b.** $P = \frac{F}{A}$
c. $P = F + A$ **d.** $P = F - A$

5. Which of the following is a conductor?
a. Glass **b.** Tungsten
c. Wood **d.** Plastic

6. What is the formula for calculating wave speed?
a. $v = \lambda f$ **b.** $v = \lambda / f$
c. $v = f - \lambda$ **d.** $v = \lambda + f$

7. Which is a greenhouse gas?
a. Argon **b.** Oxygen
c. Nitrogen **d.** Water vapor

8. What is the pressure in Pa at the bottom of a column of mercury of height 736 mm if the density of mercury is 13 530 kg/m^3?
a. $9.96 \times 10^7\ Pa$ **b.** $9.96 \times 10^3\ Pa$
c. $9.96 \times 10^5\ Pa$ **c.** $9.96 \times 10^4\ Pa$

9. What is the kinetic energy of a car of mass 1000 kg travelling at a velocity of 20 m/s?
a. $4.8kJ$ **b.** $48000kJ$
c. $48kJ$ **d.** $4800kJ$

10. Which of the following do waves transfer?
a. Matter **b.** Matter and energy
c. Energy **d.** The medium

11. If the amplitude of a wave changes, which of the following changes?
a. Wave energy **b.** Wave speed
c. Frequency **d.** Refraction

12. Which of the following is a longitudinal wave?
a. radio waves **b.** microwaves
c. water waves **d.** sound in air

13. Which of the following statement about pressure in sound waves travelling in air is correct?
a. The air particles are closest together in compression.
b. The pressure is highest in a rarefaction
c. The air particles are closest together in a rarefaction.
d. The distance between adjacent compression is equal haft a wavelength.

14. What is the effect on a water of entering shallower water?
a. Frequency increases **b.** Frequency decrease
c. Velocity decreases **d.** Velocity increase

15. In which material does sound travel fastest?
a. Water **b.** Air
c. Metal **d.** Wood

16. Which of the following is the speed of sound in air?
a. $v = d + t$ **b.** $v = d - t$
c. $v = \frac{d}{t}$ **d.** $v = \frac{2d}{t}$

17. How can speed of echo sound be defined?
a. $v = d \times t$ **b.** $v = d + t$
c. $v = \frac{d}{t}$ **d.** $v = \frac{2d}{t}$

18. Sound travels at a speed of 330 m/s. How long does it take for the sound of thunder to travel 1,485 m?
a. 45 s **b.** 4.5 s
c. 4,900 s **d.** 0.22 s

19. A boat is 5.0 m from the harbour wall. A water wave takes 6.2 s to travel from a boat to the harbour wall and back. What is the speed of the wave?
a. 0.81 m/s **b.** 1.6 m/s
c. 31 m/s **d.** 15.5 m/s

20. What is the frequency of a radio wave of wavelength 1.5 km and velocity 300 million m/s?
a. 200 kHz **b.** 20 kHz
c. 450 kHz **d.** 450 MHz

21. What is the wavelength of microwave radiation of velocity 300 million m/s and frequency 10 GHz (1GHz = 1 000 000 000 Hz)
a. 3.0 m **b.** 0.03 m
c. 33 m **d.** 33 cm

22. Which statement is the best example of heat energy transfer by conduction?
a. Heat energy is transferred from the bottom to the top of a lake.
b. Heat energy is transferred from the surface soil to the rocks below.
c. Heat energy is transferred from the Sun to the Earth.
d. Heat energy is transferred from the Earth's surface to the upper atmosphere.

23. During which process of energy exchange does cold air displace warmer air?
a. Radiation **b.** Convection
c. Absorption **d.** Conduction

B. Critical thinking question

24. What is the definition of electric power?
Ans: The electric power is the rate of the energy transfer or work done.
25. What is the difference between kinetic energy and potential energy?
Ans: Kinetic energy is energy due to motion of any objects while potential energy is the energy depends on height respected to zero condition.
26. What is the difference between elastic and inelastic collision?
Ans: Elastic collision, the kinetic is conserved. Inelastic collision, the kinetic isn't conserved.
27. What is the definition of pressure?
Ans: Pressure is the force act per unit area.
28. How many types of pressure gauge? What are they?
Ans: There are 3 types of pressure gauge. Such as bourdon gauge, u-tube manometer, mercury barometer.
29. How many modes of thermal energy transfer? What are they?
Ans: There are 3 modes of the thermal energy transfer. Such as conduction, convection and radiation.
30. What is the definition of conduction and convection?
Ans: Conduction is the transfer thermal energy from high temperature to low temperature while convection is the flow of heat through a fluid from places of higher temperature to places of lower temperature by movement of the fluid itself.
31. What is the radiation?
Ans: Radiation is the flow of heat from one place to another by means of electromagnetic waves.
32. What is the difference between conductor and insulator?
Ans: Conductor is the materials which allowed electricity or heat passes through them easily. Insulator is the materials which doesn't allow electricity or heat passes through them easily.
33. What is the difference between longitudinal wave and transverse wave?
Ans: Longitudinal wave, the particles just get vibrate parallel to the direction of the energy transfer. Transverse wave, the particles vibrate perpendicular to the direction of the energy transfer.
34. What is the difference between compression and rarefaction?
Ans: Compression is the region of high pressure. Rarefaction is the region of low pressure.
35. When the echo sound happens?
Ans: The echo sound happens when the sound wave travels bounced off the wall and reflect back through our eardrum.
36. What is the definition of period, frequency, amplitude and wavelength?
Ans: Period is time for completer one cycles.
- **Frequency is the number of waves complete one cycles.**
- **Amplitude is the distance from the undisturbed position or equilibrium.**
- **Wavelength is the distance two peaks.**

C. Work problems

37. A box exerts a force of 520 N on a floor. The bottom of the box has an area of 0.8 m^2. What is the pressure exerted by the box on the floor?
38. A triangular-based pyramid stands on a table. It has a mass of 5.5 kg, and its base has an area of 82 cm^2. Calculate the pressure (in pascals) exerted on the table by the pyramid.
39. What is the pressure 100 m below the surface of sea water of density 1150 kg/m^3?
40. A student pushes down with a force of 2 N on the plunger of syringe A. Syringe B has a plunger with an area that is 3 times the area of plunger A. What is the force produced by plunger B?
41. Calculate the power of a filament lamp operating at a voltage of 12 V and a current of 0.5 A.
42. A fixed resistor with a resistance of 25 Ω has a current of 0.8 A through it. Calculate the power of the fixed resistor.
43. (a) Calculate the resistance of a 100 W light bulb if it takes a current of 0.8 A.
(b) Calculate the power of a 12 V immersion heater with a resistance of 10 Ω. If 1kWh = $ 0.10, calculate the cost of using electricity for light bulb in a month.
44. 5 A flows through a 20 V resistor.
(a) How much electrical energy is converted to heat per second?
(b) If the current flows for one minute, how much energy is released?
45. If a note played on a guitar has a frequency of 440 Hz, what is its wavelength? (speed of sound in air = 330 m/s)
46. The A string of a guitar vibrates at 110 Hz. If the wavelength is 153 cm, what is the velocity of the wave in the string?
47. A wave in the ocean has a period of 10 s and a wavelength of 200 m. What is the wave speed?
48. There are two radio stations broadcasting on the FM radio band. One has a frequency of 101.7 MHz and the other a frequency of 100 M Hz. (speed of radio waves = 3×10^8 m/s). What is the wavelength of the radio waves from the station with the shortest wavelength?